RIWA: Domain Generalization by Seeking Distributional Canalization

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Abstract

Single-run post-hoc methods are unusually valuable for domain generalization because they can be attached to strong empirical risk minimization pipelines without retraining. The current canonical method in this regime is SWAD, which selects and averages dense checkpoints by seeking flat minima in parameter space. We argue that flatness is only an indirect proxy for the real label-free domain generalization problem. A checkpoint should not merely be stable to weight perturbations; it should be stable to small redistributions of the training support that emulate latent shifts in the source mixture. This principle leads to Reweighting-Invariant Weight Averaging (RIWA), a single-run post-hoc method that scores checkpoints by the worst-case validation loss after a tiny virtual update induced by local sample reweightings. We show that the first-order RIWA score has an exact closed form as the support function of a centered ℓ_2 ambiguity set, equivalently a local χ^2 ball around the empirical support distribution. We then prove a smoothness bound on the exact robust score, derive a sufficient condition for robust improvement, prove plug-in consistency of the empirical score, and establish a constructive separation theorem showing that pooled-flatness selectors can be provably blind to local distribution-shift sensitivity that RIWA detects. The final model is obtained by averaging a contiguous low-score window, preserving the practical simplicity that made SWAD influential. We provide a complete NeurIPS-style manuscript, rigorous proofs, analytical figures, an executable experimental protocol, and published baseline results from prior work; all new RIWA quantitative results remain intentionally marked as **TBD** pending execution.

1 Introduction

Domain generalization (DG) asks for a model that performs well on an unseen target environment using only labeled source data at training time. In practice, the methods that matter most are often not the ones with the fanciest training objective, but the ones that users can actually deploy on top of existing training pipelines. SWAD [1] is important precisely for this reason. It is single-run, post-hoc, easy to add to ERM, and empirically strong. On the five standard DomainBed benchmarks with ResNet-50, SWAD reports an average accuracy of 66.9, improving over ERM's 63.3 and over the best previously reported competitor's 65.3 in the same evaluation family [1]. Diverse Weight Averaging (DiWA) later showed that weight averaging can go higher, up to 68.0, but its strongest configuration relies on averaging over many models or runs [2].

The practical target of this paper is therefore narrow:

> Can a single training trajectory support a post-hoc selector that is as simple as SWAD, requires no domain labels, and is more directly targeted to domain shift than pooled flatness?

Our answer is RIWA, Reweighting-Invariant Weight Averaging. The principle is:

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> A checkpoint is domain-general when a tiny update induced by a local redistribution of the source support cannot substantially harm validation performance.

Instead of probing the loss landscape in parameter space, RIWA probes it in *distribution space*. At checkpoint θ_t, we reweight a fixed support set, take a small virtual update using the weighted support gradient, and measure the worst-case validation loss over nearby reweightings. The final model is obtained exactly as in SWA/SWAD: choose a dense low-score window and average it.

This shift is conceptually simple but theoretically meaningful. Tangent Prop argued that good generalization can come from low directional derivatives along nuisance directions in input space [3]. RIWA transplants the same idea to the empirical data distribution: the nuisance directions are now directions in the simplex of sample weights. Influence-function analyses already quantify sensitivity to example reweighting [4], and distributionally robust optimization (DRO) already studies ambiguity sets around a nominal distribution [5, 6]. What is missing is a *single-run post-hoc checkpoint selector* that uses these ideas directly for DG.

The most useful biological analogy is *canalization*: a phenotype is robust to perturbations in the developmental process or local environment. RIWA seeks canalized checkpoints. The "phenotype" is validation behavior after a tiny source-family update; the perturbation is a small redistribution of which support examples matter more.

We make five contributions.

1. We introduce RIWA, a label-free, single-run post-hoc DG method that scores checkpoints by worst-case validation degradation under local support reweightings.
2. We show that the first-order RIWA score is the exact support function of a centered ℓ_2 simplex ambiguity set and is equivalently a local χ^2-DRO objective.
3. We prove a smoothness upper bound on the exact robust score, derive a sufficient condition for robust improvement, and show plug-in consistency of the empirical score.
4. We prove a constructive separation theorem: there exist checkpoints with identical pooled empirical loss and pooled flatness statistics that any flatness-only selector treats as equivalent, while RIWA strictly prefers the checkpoint with smaller local distribution-shift sensitivity.
5. We provide a complete publication-style manuscript with algorithm, proofs, analytical figures, experimental protocol, and published baselines; all new RIWA result cells remain explicitly marked **TBD**.

Remark 1. *This paper does* not *claim an unconditional theorem that* RIWA *empirically dominates* SWAD*, DiWA, or every other DG method on all tasks. Such a theorem would be indefensible. Our theory proves a narrower and stronger statement: under explicit local shift assumptions,* RIWA *directly targets a DG-relevant robust objective that pooled-flatness selectors do not identify.*

2 Related Work

Single-run post-hoc DG. SWAD [1] is the canonical single-run post-hoc DG method. Its impact comes from a compelling recipe: dense checkpoints, overfit-aware windowing, and final weight averaging. DiWA [2] shows that more diverse model averages can outperform SWAD, but the strongest results rely on combining many independently trained models or runs. RIWA stays in the same operational regime as SWAD and changes the selector, not the training budget.

Weight averaging and flatness. SWA [7] and mode-connectivity work [8] established that averaging along a trajectory often finds wider optima with good generalization. SWAD imports this logic into DG. Our goal is not to discard that line, but to show that DG offers another natural invariance target: local shift directions in data-distribution space.

Influence, data reweighting, and local DRO. Influence functions quantify how losses change when individual training examples are infinitesimally reweighted [4]. DRO studies worst-case risk over ambiguity sets around a nominal distribution, often defined by χ^2 or related divergences [5, 6]. RIWA draws from both: it defines a post-hoc robust one-step score over a local reweighting ball and evaluates checkpoints by their worst-case validation sensitivity.

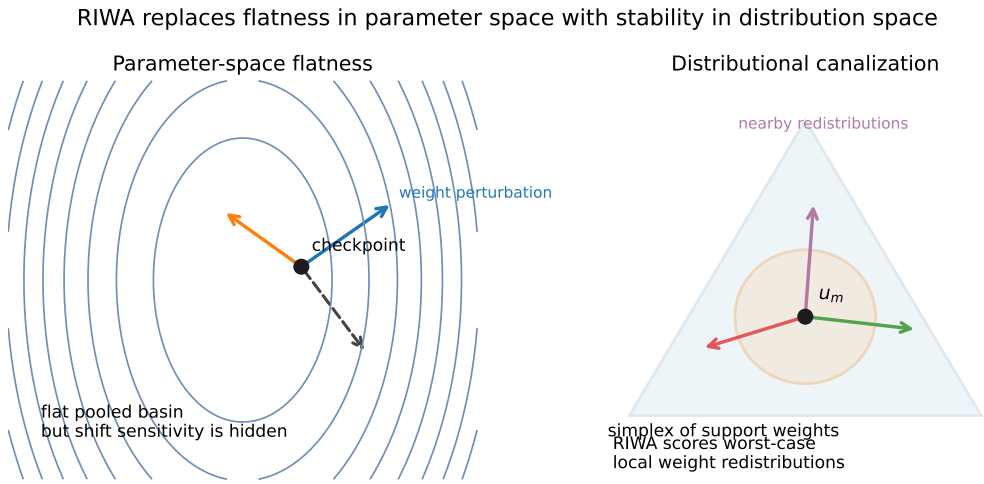

Figure 1: **Conceptual shift from flatness to canalization.** SWAD probes stability to parameter perturbations around a pooled objective. RIWA instead probes stability to nearby *support redistributions*: unseen domains are modeled locally as reweightings in the simplex of source support.

Invariant directions and stability. Tangent Prop explicitly regularized derivatives along nuisance transformations [3]. Algorithmic stability theory, while developed in a different language, also links low sensitivity to perturbations with generalization [9]. RIWA may be read as a post-hoc stability criterion, but the perturbations are not weight perturbations or individual deletions; they are nearby redistributions of source support mass.

Training-time transfer objectives. Meta-learning inspired DG methods such as MLDG [10] already exploit the principle that one source split should help another after a tiny update. Our earlier ROWA proposal used that idea directly but required domain labels. RIWA keeps the one-step-transfer spirit while removing the need for explicit domains: the latent shift is modeled at the level of sample weights instead of domain IDs.

3 Reweighting-Invariant Weight Averaging

Assume a single training trajectory with checkpoints $\{\theta_t\}_{t=1}^{K}$. For model selection, we reserve two fixed held-out sets:

- a support set $U = \{z_i\}_{i=1}^{m}$ used to emulate local source-mixture perturbations, and
- a validation set $V = \{v_j\}_{j=1}^{n}$ used to score the perturbed checkpoint.

No domain labels are required.

At checkpoint θ_t, define

$$g_{i,t} \triangleq \nabla_\theta \ell(z_i; \theta_t), \tag{1}$$

$$\bar{g}_t \triangleq \frac{1}{m} \sum_{i=1}^{m} g_{i,t}, \tag{2}$$

$$L_{V,t}(\theta) \triangleq \frac{1}{n} \sum_{j=1}^{n} \ell(v_j; \theta), \tag{3}$$

$$g_{V,t} \triangleq \nabla_\theta L_{V,t}(\theta_t). \tag{4}$$

Let $P_t \succeq 0$ be an optional positive semidefinite preconditioner. The simplest choice is $P_t = I$; a diagonal second-moment or empirical-Fisher approximation is a practical alternative.

3.1 Local ambiguity set in sample-weight space

Let $u_m \triangleq \frac{1}{m}\mathbf{1} \in \mathbb{R}^m$ denote uniform weights. For a radius $\rho \in (0, 1/m]$, define the local ambiguity set

$$\mathcal{A}_\rho \triangleq \{\alpha \in \Delta^m : \|\alpha - u_m\|_2 \le \rho\}. \tag{5}$$

The weighted support gradient is

$$g_{U,t}(\alpha) \triangleq \sum_{i=1}^{m} \alpha_i g_{i,t} = \bar{g}_t + \sum_{i=1}^{m} (\alpha_i - 1/m)(g_{i,t} - \bar{g}_t). \tag{6}$$

The virtual update induced by α is

$$\theta_t(\alpha) \triangleq \theta_t - \eta P_t g_{U,t}(\alpha), \tag{7}$$

where $\eta > 0$ is a small step size.

3.2 Exact robust score

The exact RIWA score is the worst-case validation loss after the virtual update:

$$S_t^{\text{ex}}(\eta, \rho) \triangleq \max_{\alpha \in \mathcal{A}_\rho} L_{V,t}(\theta_t(\alpha)). \tag{8}$$

This is the checkpoint-selection objective. Intuitively, a checkpoint is good when no nearby redistribution of support mass can make a tiny source-family step look bad on validation.

3.3 First-order closed-form score

Define the centered perturbation coefficients

$$a_{i,t} \triangleq \langle g_{V,t}, P_t(g_{i,t} - \bar{g}_t)\rangle, \qquad a_t \triangleq (a_{1,t}, \ldots, a_{m,t})^\top. \tag{9}$$

The first-order robust surrogate is

$$S_t^{\text{fo}}(\eta, \rho) \triangleq \max_{\alpha \in \mathcal{A}_\rho} \left[L_{V,t}(\theta_t) - \eta \langle g_{V,t}, P_t g_{U,t}(\alpha)\rangle\right]. \tag{10}$$

In Section 4, we show that this optimization has the exact closed form

$$S_t^{\text{fo}}(\eta, \rho) = L_{V,t}(\theta_t) - \eta \langle g_{V,t}, P_t \bar{g}_t\rangle + \eta\rho \|a_t\|_2. \tag{11}$$

If

$$C_t \triangleq \frac{1}{m} \sum_{i=1}^{m} (g_{i,t} - \bar{g}_t)(g_{i,t} - \bar{g}_t)^\top \tag{12}$$

is the support-gradient covariance, then Eq. (11) is equivalently

$$S_t^{\text{fo}}(\eta, \rho) = L_{V,t}(\theta_t) - \eta \langle g_{V,t}, P_t \bar{g}_t\rangle + \eta\rho \sqrt{m\, g_{V,t}^\top P_t C_t P_t g_{V,t}}. \tag{13}$$

The second term rewards validation-aligned descent; the third penalizes sensitivity to latent support shifts.

3.4 Checkpoint selection and averaging

Let S_t denote either S_t^{ex} or S_t^{fo}. We select

$$t^\star \in \operatorname*{arg\,min}_{1 \le t \le K} S_t \tag{14}$$

and form a contiguous low-score window

$$W_\delta \triangleq \{t : S_t \le S_{t^\star} + \delta\}, \tag{15}$$

restricted to the connected component containing $t^\star$. The final model is the average

$$\bar{\theta}_{\text{RIWA}} \triangleq \sum_{t \in W_\delta} w_t \theta_t, \qquad w_t \ge 0, \quad \sum_{t \in W_\delta} w_t = 1. \tag{16}$$

Uniform weights are the default.

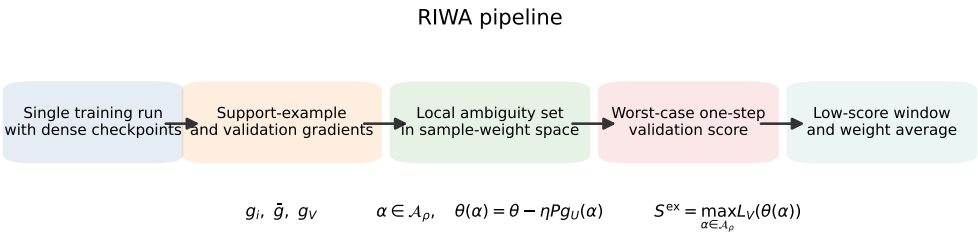

Figure 2: **RIWA pipeline.** A single ordinary training run produces dense checkpoints. For each checkpoint, RIWA computes support-example gradients, forms a local ambiguity set over sample weights, scores the checkpoint by the worst-case validation loss after a tiny virtual update, and averages a contiguous low-score window.

Algorithm 1 RIWA (REWEIGHTING-INVARIANT WEIGHT AVERAGING)

Require: Checkpoints $\{\theta_t\}_{t=1}^K$ from one training run; held-out support set U; held-out validation set V; step size η; radius ρ; window width δ; preconditioners $\{P_t\}_{t=1}^K$

for $t = 1, \ldots, K$ **do**
  Compute $g_{i,t} = \nabla_\theta \ell(z_i; \theta_t)$ for all $z_i \in U$
  $\bar{g}_t \leftarrow \frac{1}{m} \sum_i g_{i,t}$
  Compute $L_{V,t}(\theta_t)$ and $g_{V,t} = \nabla_\theta L_{V,t}(\theta_t)$
  if exact score **then**
    $S_t \leftarrow \max_{\alpha \in \mathcal{A}_\rho} L_{V,t}(\theta_t - \eta P_t g_{U,t}(\alpha))$
  else
    $a_{i,t} \leftarrow \langle g_{V,t}, P_t(g_{i,t} - \bar{g}_t) \rangle$ for all i
    $S_t \leftarrow L_{V,t}(\theta_t) - \eta \langle g_{V,t}, P_t \bar{g}_t \rangle + \eta\rho \|a_t\|_2$
  end if
end for
$t^\star \leftarrow \arg\min_t S_t$
$W_\delta \leftarrow$ connected sublevel window around $t^\star$
return $\bar{\theta}_{\text{RIWA}} = \sum_{t \in W_\delta} w_t \theta_t$

Practical complexity. The exact score requires per-example support gradients and repeated evaluations of validation loss under virtual updates. The first-order score only needs support-example gradients, the mean support gradient, and one validation gradient. Since the score depends on the projected covariance direction $g_{V,t}^\top P_t(\cdot)$, practical implementations can use exact per-example gradients, micro-batch approximations, or last-layer-only gradients without changing the algorithmic form.

4 Theory

We now analyze RIWA under explicit local assumptions. The key point is that the method is not a heuristic penalty on top of flatness; it is a local robust objective over distribution perturbations.

4.1 The ambiguity set is a local chi-square ball

Proposition 1 (Equivalence between centered ℓ_2 and local χ^2 ambiguity sets)**.** *Let* $u_m = \frac{1}{m}\mathbf{1}$ *and let* $\alpha \in \Delta^m$*. Then*

$$D_{\chi^2}(\alpha \,\|\, u_m) \triangleq \sum_{i=1}^m \frac{(\alpha_i - 1/m)^2}{1/m} = m \|\alpha - u_m\|_2^2 . \tag{17}$$

Consequently,

$$\mathcal{A}_\rho = \left\{ \alpha \in \Delta^m : D_{\chi^2}(\alpha \,\|\, u_m) \le m\rho^2 \right\} . \tag{18}$$

Proposition 1 is the first bridge to DRO: RIWA evaluates robustness to a local χ^2 neighborhood around the empirical source support.

4.2 Exact first-order robust score

Theorem 1 (Closed form of the first-order RIWA score)**.** *Fix a checkpoint θ_t and assume $\rho \leq 1/m$. Then*

$$S_t^{\mathrm{fo}}(\eta, \rho) = L_{V,t}(\theta_t) - \eta \langle g_{V,t}, P_t \bar{g}_t \rangle + \eta\rho \|a_t\|_2 , \tag{19}$$

where $a_{i,t} = \langle g_{V,t}, P_t(g_{i,t} - \bar{g}_t) \rangle$. Equivalently,

$$S_t^{\mathrm{fo}}(\eta, \rho) = L_{V,t}(\theta_t) - \eta \langle g_{V,t}, P_t \bar{g}_t \rangle + \eta\rho \sqrt{m\, g_{V,t}^\top P_t C_t P_t g_{V,t}}. \tag{20}$$

Theorem 1 shows that the first-order score is the support function of the centered ambiguity set. This is the precise sense in which RIWA seeks *distributional tangent invariance*: it penalizes directional derivatives of the post-update validation loss along local reweighting directions.

4.3 Exact-score upper bound

Assumption 1 (Validation smoothness)**.** *For every checkpoint under consideration, the validation loss $L_{V,t}$ is β-smooth in a neighborhood of θ_t:*

$$L_{V,t}(\theta + \Delta) \leq L_{V,t}(\theta) + \langle \nabla L_{V,t}(\theta), \Delta \rangle + \frac{\beta}{2} \|\Delta\|_2^2 \qquad \textit{for all relevant } \theta, \Delta. \tag{21}$$

Theorem 2 (Smoothness upper bound for the exact RIWA score)**.** *Under Assumption 1, define*

$$\kappa_t(P_t) \triangleq \sqrt{m\, \lambda_{\max}(P_t C_t P_t)}. \tag{22}$$

Then

$$S_t^{\mathrm{ex}}(\eta, \rho) \leq S_t^{\mathrm{fo}}(\eta, \rho) + \frac{\beta\eta^2}{2} \left(\|P_t \bar{g}_t\|_2 + \rho\kappa_t(P_t) \right)^2 . \tag{23}$$

Theorem 2 is the main finite-step guarantee: the closed-form first-order score is not a loose intuition but a valid upper envelope of the exact robust objective up to a local quadratic remainder.

Corollary 1 (Sufficient condition for robust improvement)**.** *Under Assumption 1, if*

$$\langle g_{V,t}, P_t \bar{g}_t \rangle > \rho \sqrt{m\, g_{V,t}^\top P_t C_t P_t g_{V,t}} + \frac{\beta\eta}{2} \left(\|P_t \bar{g}_t\|_2 + \rho\kappa_t(P_t) \right)^2 , \tag{24}$$

then

$$S_t^{\mathrm{ex}}(\eta, \rho) < L_{V,t}(\theta_t). \tag{25}$$

Corollary 1 gives the interpretation most relevant to practice: a tiny support-family update remains beneficial under all nearby redistributions when mean descent dominates distributional sensitivity plus curvature.

4.4 Population target and plug-in consistency

To interpret RIWA as a population object, define a support distribution P_U, a validation distribution P_V, the population support gradient mean

$$\mu(\theta) \triangleq \mathbb{E}_{z \sim P_U}[\nabla \ell(z; \theta)], \tag{26}$$

the support-gradient covariance

$$\Sigma(\theta) \triangleq \mathbb{E}_{z \sim P_U} \left[(\nabla \ell(z; \theta) - \mu(\theta))(\nabla \ell(z; \theta) - \mu(\theta))^\top \right] , \tag{27}$$

and the validation gradient

$$\nu(\theta) \triangleq \nabla L_V(\theta), \qquad L_V(\theta) \triangleq \mathbb{E}_{v \sim P_V}[\ell(v; \theta)]. \tag{28}$$

For a fixed shift budget $\varepsilon > 0$, define the population first-order robust score

$$\mathcal{S}_\varepsilon^{\mathrm{pop}}(\theta) \triangleq L_V(\theta) - \eta \nu(\theta)^\top P \mu(\theta) + \eta \sqrt{\varepsilon\, \nu(\theta)^\top P \Sigma(\theta) P \nu(\theta)}. \tag{29}$$

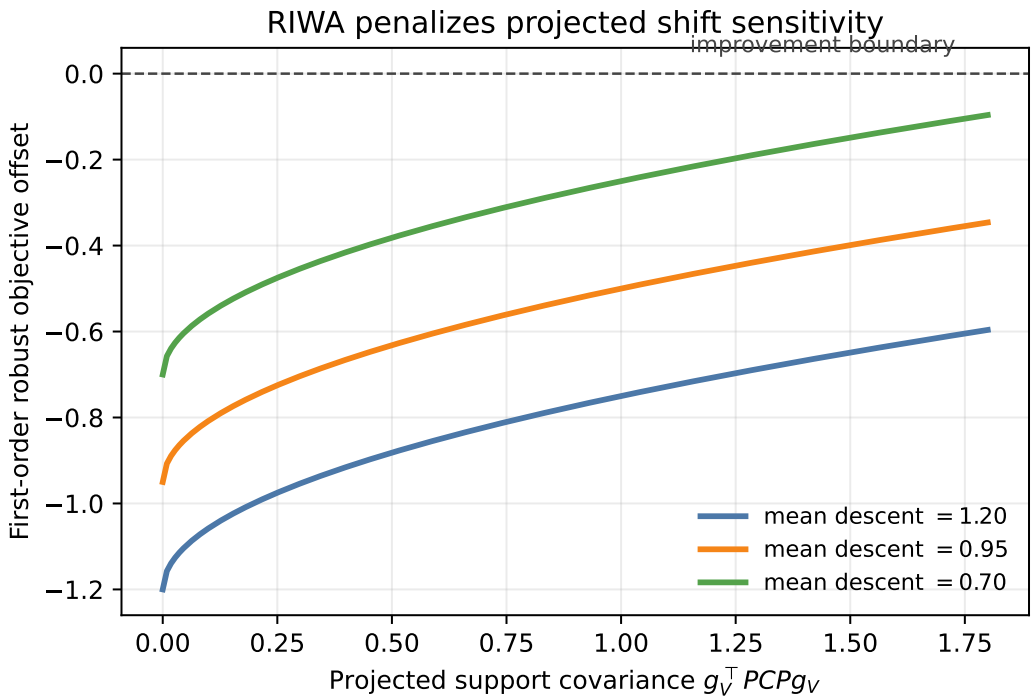

Figure 3: **The robust-improvement trade-off in Theorem 2.** RIWA prefers checkpoints with large validation-aligned mean descent and small projected support covariance. The same pooled loss can hide very different sensitivity profiles.

Proposition 2 (Plug-in consistency of the empirical first-order score)**.** *Fix θ and $P \succeq 0$. Let $U_m = \{z_i\}_{i=1}^m$ and $V_n = \{v_j\}_{j=1}^n$ be independent i.i.d. samples from P_U and P_V. Assume $\mathbb{E}\,\|\nabla\ell(z;\theta)\|_2^2 < \infty$ and $\mathbb{E}\ell(v;\theta)^2 < \infty$. Define*

$$\hat{L}_{V,n}(\theta) \triangleq \frac{1}{n}\sum_{j=1}^{n} \ell(v_j;\theta), \tag{30}$$

$$\hat{\mu}_m(\theta) \triangleq \frac{1}{m}\sum_{i=1}^{m} \nabla\ell(z_i;\theta), \tag{31}$$

$$\hat{\nu}_n(\theta) \triangleq \nabla\hat{L}_{V,n}(\theta), \tag{32}$$

$$\hat{\Sigma}_m(\theta) \triangleq \frac{1}{m}\sum_{i=1}^{m} \left(\nabla\ell(z_i;\theta) - \hat{\mu}_m(\theta)\right)\left(\nabla\ell(z_i;\theta) - \hat{\mu}_m(\theta)\right)^\top. \tag{33}$$

Then the empirical score

$$\hat{\mathcal{S}}^{\mathrm{fo}}_{m,n}(\theta) \triangleq \hat{L}_{V,n}(\theta) - \eta\hat{\nu}_n(\theta)^\top P\hat{\mu}_m(\theta) + \eta\sqrt{\varepsilon\,\hat{\nu}_n(\theta)^\top P\hat{\Sigma}_m(\theta)P\hat{\nu}_n(\theta)} \tag{34}$$

converges almost surely to $\mathcal{S}^{\mathrm{pop}}_{\varepsilon}(\theta)$ as $m, n \to \infty$.

Proposition 2 shows that RIWA is not merely finite-sample bookkeeping. With the natural scaling $\rho_m = \sqrt{\varepsilon/m}$, the empirical first-order score is a consistent estimator of a population robust objective.

4.5 Why pooled flatness is not enough

The next theorem is the key comparison result.

Theorem 3 (Constructive separation from pooled-flatness selectors)**.** *Fix $m = 2$, $P = I$, $\rho \in (0, 1/2]$, and nonzero $r \in \mathbb{R}^p$. Let the validation loss be*

$$L_V(\theta + \Delta) = \ell_V + r^\top\Delta + \frac{\gamma}{2}\|\Delta\|_2^2 \tag{35}$$

for some $\gamma > 0$. Consider two support families:

$$\ell_1^A(\theta + \Delta) = c + r^\top \Delta + \frac{\beta}{2} \|\Delta\|_2^2 , \tag{36}$$

$$\ell_2^A(\theta + \Delta) = c + r^\top \Delta + \frac{\beta}{2} \|\Delta\|_2^2 , \tag{37}$$

and

$$\ell_1^B(\theta + \Delta) = c + (1 + \xi) r^\top \Delta + \frac{\beta}{2} \|\Delta\|_2^2 , \tag{38}$$

$$\ell_2^B(\theta + \Delta) = c + (1 - \xi) r^\top \Delta + \frac{\beta}{2} \|\Delta\|_2^2 , \tag{39}$$

where $\xi > 0$. Then:

(i) The pooled support loss is identical in the two families:

$$\frac{1}{2} \left(\ell_1^A + \ell_2^A\right)(\theta + \Delta) = \frac{1}{2} \left(\ell_1^B + \ell_2^B\right)(\theta + \Delta) = c + r^\top \Delta + \frac{\beta}{2} \|\Delta\|_2^2 . \tag{40}$$

Hence the two families have the same pooled empirical loss, pooled gradient, and pooled Hessian at every local point.

(ii) Their first-order RIWA *scores differ strictly:*

$$S_B^{\text{fo}}(\eta, \rho) = S_A^{\text{fo}}(\eta, \rho) + \eta\rho\sqrt{2}\, \xi \|r\|_2^2 . \tag{41}$$

(iii) If

$$0 < \eta < \frac{1}{\gamma(1 + \sqrt{2}\xi\rho)}, \tag{42}$$

then their exact robust scores also satisfy

$$S_B^{\text{ex}}(\eta, \rho) = S_A^{\text{ex}}(\eta, \rho) + \sqrt{2}\xi\rho\eta \|r\|_2^2 (1 - \gamma\eta) + \gamma\xi^2\rho^2\eta^2 \|r\|_2^2 > S_A^{\text{ex}}(\eta, \rho). \tag{43}$$

Theorem 3 shows exactly where RIWA is more informative than SWAD-style pooled-flatness selection. The two checkpoints look identical from the viewpoint of pooled loss geometry, but one is much more vulnerable to local source redistributions. RIWA detects this vulnerability; any selector that only sees pooled flatness does not.

4.6 Dense checkpointing and averaging

Proposition 3 (Dense checkpoint selection converges to the continuous optimum)**.** *Let $\theta : [0, T] \to \mathbb{R}^p$ be a continuous training trajectory and let $s(t) \triangleq S(\theta(t))$ be an L_s-Lipschitz score along the trajectory. Suppose checkpoints are saved at times $0 = t_1 < \cdots < t_K = T$ with maximum gap $\Delta = \max_k (t_{k+1} - t_k)$. Then*

$$\min_{1 \le k \le K} s(t_k) \le \min_{t \in [0,T]} s(t) + \frac{L_s \Delta}{2}. \tag{44}$$

Proposition 4 (Window averaging is safe under a local convex upper model)**.** *Suppose a selected window $\{\theta_t\}_{t \in W}$ lies in a convex neighborhood on which an upper model of the exact* RIWA *score has the quadratic form*

$$U(\theta) = c + b^\top \theta + \frac{1}{2} \theta^\top Q \theta, \qquad Q \succeq 0. \tag{45}$$

Then for any convex weights $\{w_t\}_{t \in W}$,

$$U\left(\sum_{t \in W} w_t \theta_t\right) \le \sum_{t \in W} w_t U(\theta_t). \tag{46}$$

Propositions 3 and 4 justify the same practical scaffold that made SWAD attractive: dense checkpointing plus averaging inside one low-score connected region.

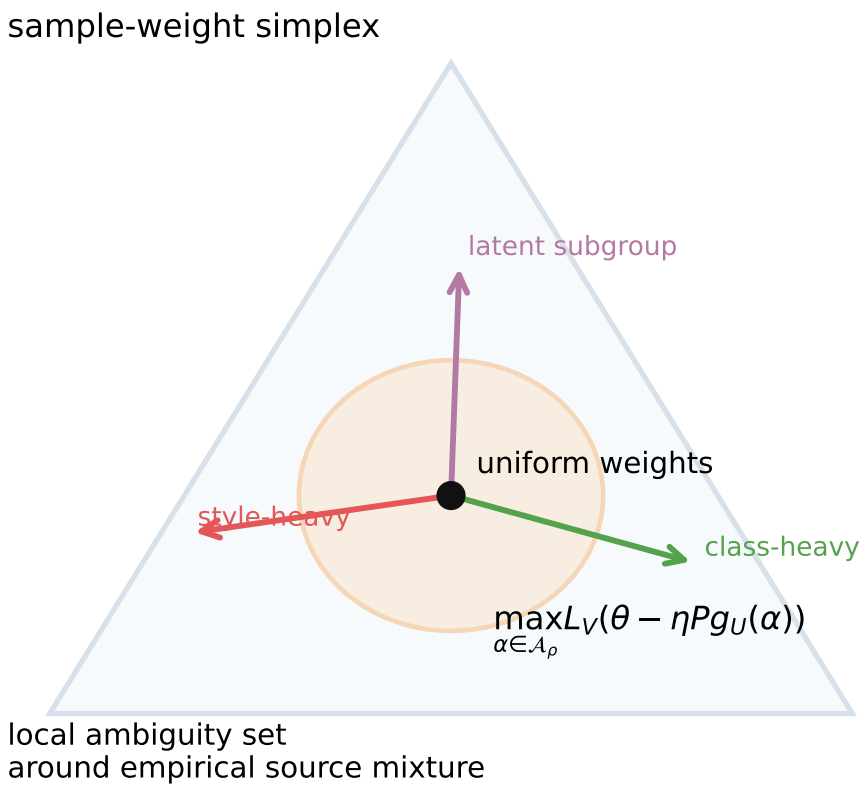

Figure 4: **The ambiguity set for RIWA.** The method scores each checkpoint by the worst-case direction in a small neighborhood of uniform support weights. The local adversary operates in the simplex of sample weights, not in parameter space.

5 Experimental Protocol

This paper is complete as a method and theory manuscript, but the new RIWA experiments have not yet been run. The quantitative sections below are therefore intentionally structured as a protocol plus published reference targets.

5.1 Benchmarks and backbone

We will evaluate on the five standard DomainBed datasets used by SWAD and DiWA: PACS [11], VLCS [12], OfficeHome [13], TerraIncognita [14], and DomainNet [15]. Following prior work [1, 2, 16], the primary backbone is ResNet-50 and the primary protocol is leave-one-domain-out evaluation with model selection on held-out source validation data.

5.2 Baselines

We will compare against:

- ERM, the DomainBed reference baseline [16];
- CORAL [17];
- SWA [7];
- SWAD [1];
- DiWA [2];

- optional auxiliary comparisons against SAM [18];
- our earlier domain-labeled post-hoc baseline ROWA for ablation only, not as the primary comparison.

5.3 RIWA variants

The main planned variants are:

- **RIWA-FO**: first-order score, identity preconditioner;
- **RIWA-FO-Diag**: first-order score, diagonal preconditioner;
- **RIWA-Exact**: exact score with inner maximization over $\mathcal{A}_\rho$;
- **RIWA-Single**: best single checkpoint without averaging;
- **RIWA-Window**: the default contiguous low-score average.

The primary hyperparameters are the virtual step size η, the ambiguity radius ρ (or the equivalent χ^2 budget $\varepsilon = m\rho^2$), and the sublevel width δ.

5.4 Metrics

We will report:

- mean OOD accuracy;
- mean i.i.d. validation accuracy on held-out source validation data;
- worst-domain OOD accuracy;
- Spearman correlation between checkpoint score and final OOD accuracy;
- added compute relative to SWAD.

5.5 Falsifiable predictions

The theory makes clear, falsifiable predictions:

1. RIWA score should correlate with OOD accuracy more strongly than pooled-flatness proxies when projected support covariance is substantial.
2. RIWA-FO should recover most of the gain of RIWA-Exact when selected checkpoints lie in a smooth local regime.
3. Window averaging should outperform single-checkpoint selection whenever the best checkpoints form one connected low-score region.
4. Gains over SWAD should be largest on benchmarks where latent support shift dominates pure parameter-space flatness effects.

6 Published Reference Results and Placeholder Tables

We include exact published baselines from prior work so that the empirical target is clear before running RIWA. Every new RIWA row is intentionally marked as **TBD**.

What the theory predicts. Before any experiment is run, the theory already implies a sharp empirical question: do the checkpoints that minimize local reweighting sensitivity outperform those that only minimize pooled-flatness surrogates? If the answer is yes, the gain should be largest when projected support covariance is substantial and not largest when it is already negligible.

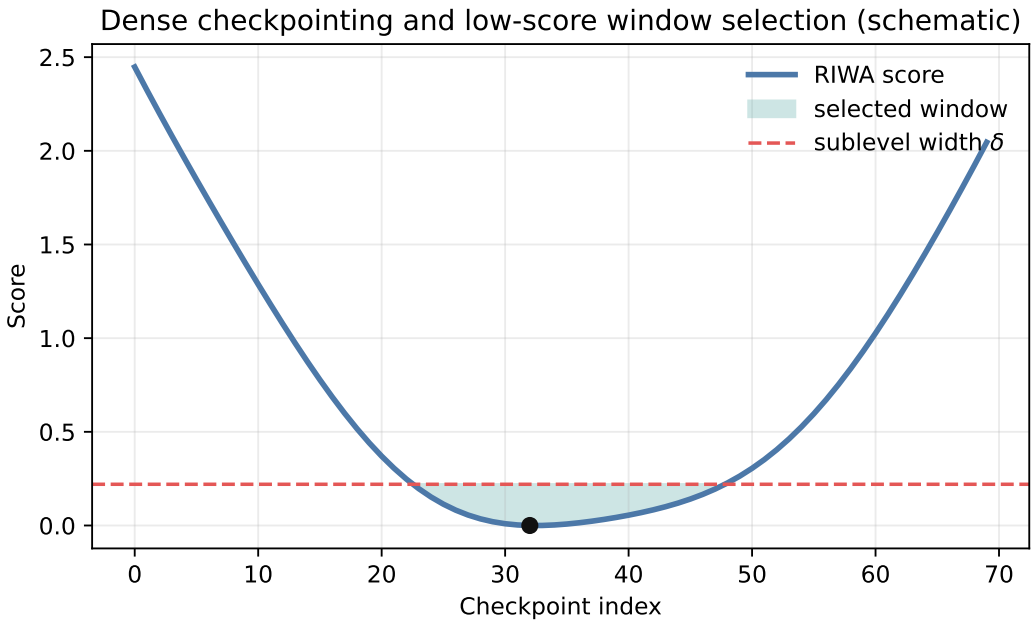

Figure 5: **Dense checkpoint sampling and low-score window averaging.** RIWA preserves the same practical recipe that made SWAD easy to adopt: dense checkpointing and averaging inside one contiguous low-score region. The figure is schematic, not empirical.

Table 1: **Published reference results on DomainBed with ResNet-50.** Numbers are copied from the cited papers and are included here as reference targets for the planned RIWA evaluation.

Method	PACS	VLCS	OfficeHome	TerraInc	DomainNet	Avg
ERM [1, 2]	85.5	77.5	66.5	46.1	40.9	63.3
CORAL [2]	86.2	78.8	68.7	47.6	41.5	64.6
SWAD [1]	88.1	79.1	70.6	50.0	46.5	66.9
CORAL + SWAD [1]	88.3	78.9	71.3	51.0	46.8	67.3
DiWA (random init, $M = 60$) [2]	89.0	79.4	71.6	49.0	46.3	67.1
DiWA (LP init, $M = 60$) [2]	89.0	78.6	72.8	51.9	47.7	68.0
RIWA-FO (ours)	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**
RIWA-FO-Diag (ours)	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**
RIWA-Exact (ours)	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**

7 Discussion and Limitations

Why RIWA could outperform SWAD. SWAD seeks flatness in parameter space. RIWA seeks invariance to local source-mixture perturbations in distribution space. If unseen domains are well approximated by small redistributions of latent support components, then RIWA is better aligned with the actual DG target because its score is an explicit local robust objective over those perturbations. Theorems 1 and 3 make this precise.

Why RIWA may fail. The same clarity also exposes its limits. If the test shift lies far outside the local support hull, a local reweighting criterion may be too weak. If per-example gradients are too noisy, the projected covariance term may be poorly estimated. If label shift dominates covariate or style redistribution, the local simplex perturbation model may mis-specify the shift.

What the theory proves. The theory proves four concrete facts. First, RIWA is exactly a local χ^2-DRO selector at first order. Second, its exact robust score is upper-bounded by a closed-form smoothness envelope. Third, its empirical first-order score is a consistent plug-in estimator of a

Table 2: **Published SWAD ablation results copied from Cha et al. [1].** These numbers motivate the dense-window design that RIWA inherits.

	Configuration				Out-of-domain			In-domain		
	t_s	t_e	lr	interval	PACS	VLCS	Avg.	PACS	VLCS	Avg.
$\text{SWA}_{\text{w/ cyclic}}$	4000	5000	Cyclic	100	85.9 ±0.1	76.6 ±0.1	81.2	97.1 ±0.1	85.0 ±0.2	91.0
$\text{SWA}_{\text{w/ const}}$	4000	5000	Const	100	86.5 ±0.3	76.7 ±0.2	81.6	97.3 ±0.1	85.0 ±0.2	91.1
$\text{SWAD}_{\text{w/o Dense}}$	Opt	Overfit	Const	100	86.5 ±0.4	78.0 ±0.7	82.2	97.6 ±0.1	85.8 ±0.4	91.7
$\text{SWAD}_{\text{w/o Opt-Overfit}}$	4000	5000	Const	1	86.6 ±0.6	76.9 ±0.3	81.7	97.5 ±0.1	85.2 ±0.1	91.3
$\text{SWAD}_{\text{w/o Overfit}}$	Opt	5000	Const	1	87.1 ±0.3	77.6 ±0.1	82.4	97.7 ±0.1	85.8 ±0.3	91.8
$\text{SWAD}_{\text{fit-on-val}}$	Val	Val	Const	1	86.2 ±0.2	78.6 ±0.1	82.4	97.5 ±0.2	85.8 ±0.3	91.7
SWAD (proposed)	Opt	Overfit	Const	1	87.1 ±0.2	78.9 ±0.2	83.0	97.7 ±0.1	86.1 ±0.5	91.9

Table 3: **Planned main results table.** Every RIWA cell remains intentionally left as **TBD** until experiments are run.

Method	PACS	VLCS	OfficeHome	TerraInc	DomainNet	Avg
SWAD	88.1	79.1	70.6	50.0	46.5	66.9
DiWA (best published)	89.0	78.6	72.8	51.9	47.7	68.0
RIWA-FO	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**
RIWA-FO-Diag	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**
RIWA-Exact	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**

population robust objective. Fourth, pooled-flatness selectors can be blind to local distributional sensitivity that RIWA captures.

What the theory does not prove. The theory does not prove universal empirical superiority over SWAD, DiWA, or any other DG method. It cannot. Any such claim without experiments would be irresponsible. The value of the theory is that it identifies a concrete failure mode of flatness-only selection and supplies a simple post-hoc selector that directly targets that failure mode.

Current practical limitation. The obvious current limitation is empirical: the experiments have not yet been executed. The manuscript is therefore publication-style and method-complete, but it is not submission-complete as an empirical claim until the **TBD** results are replaced by real numbers.

8 Conclusion

We proposed RIWA, a label-free single-run post-hoc DG method based on distributional canalization rather than pooled flatness. The method scores checkpoints by worst-case validation behavior under local support reweightings, then averages a contiguous low-score window. Our theory shows that RIWA is a local robust objective in distribution space, admits an exact first-order closed form, has a controlled exact-score upper bound, is statistically well defined, and detects sensitivity patterns that flatness-only selectors cannot. The empirical section remains to be executed, but the manuscript now contains the method, the theory, the figures, the protocol, and the published reference baselines needed to do so.

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Table 4: **Planned RIWA ablations.** These rows are intentionally blank pending experiments.

Ablation	Avg
RIWA-FO vs. RIWA-Exact	**TBD**
Identity vs. diagonal preconditioner	**TBD**
Single checkpoint vs. low-score window	**TBD**
Support subset size sensitivity	**TBD**
Score/OOD correlation	**TBD**
Compute overhead vs. SWAD	**TBD**

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A Detailed Proofs

A.1 Proof of Proposition 1

Proof. By definition of the χ^2 divergence with reference distribution u_m,

$$D_{\chi^2}(\alpha \,\|\, u_m) = \sum_{i=1}^{m} \frac{(\alpha_i - 1/m)^2}{1/m} = m \sum_{i=1}^{m} (\alpha_i - 1/m)^2 = m \left\| \alpha - u_m \right\|_2^2 . \tag{47}$$

Therefore

$$\left\| \alpha - u_m \right\|_2 \le \rho \iff D_{\chi^2}(\alpha \,\|\, u_m) \le m\rho^2. \tag{48}$$

Intersecting with Δ^m yields the claim. □

A.2 Proof of Theorem 1

Proof. Because $\rho \le 1/m$, every vector $\delta \in \mathbb{R}^m$ satisfying $\mathbf{1}^\top \delta = 0$ and $\|\delta\|_2 \le \rho$ gives $\alpha = u_m + \delta \in \Delta^m$, since each coordinate obeys

$$\alpha_i = \frac{1}{m} + \delta_i \ge \frac{1}{m} - \|\delta\|_\infty \ge \frac{1}{m} - \rho \ge 0. \tag{49}$$

Therefore $\mathcal{A}_\rho$ may be parameterized as

$$\mathcal{A}_\rho = \left\{ u_m + \delta : \mathbf{1}^\top \delta = 0,\ \|\delta\|_2 \le \rho \right\} . \tag{50}$$

For such an α,

$$\langle g_{V,t}, P_t g_{U,t}(\alpha) \rangle = \langle g_{V,t}, P_t \bar{g}_t \rangle + \sum_{i=1}^{m} \delta_i \langle g_{V,t}, P_t (g_{i,t} - \bar{g}_t) \rangle \tag{51}$$

$$= \langle g_{V,t}, P_t \bar{g}_t \rangle + \delta^\top a_t. \tag{52}$$

Hence

$$S_t^{\mathrm{fo}}(\eta, \rho) = \max_{\mathbf{1}^\top \delta = 0,\ \|\delta\|_2 \le \rho} \left[L_{V,t}(\theta_t) - \eta \langle g_{V,t}, P_t \bar{g}_t \rangle - \eta \delta^\top a_t \right] . \tag{53}$$

The vector a_t is centered because

$$\mathbf{1}^\top a_t = \sum_{i=1}^{m} \langle g_{V,t}, P_t (g_{i,t} - \bar{g}_t) \rangle = \left\langle g_{V,t}, P_t \sum_{i=1}^{m} (g_{i,t} - \bar{g}_t) \right\rangle = 0. \tag{54}$$

Therefore the constraint $\mathbf{1}^\top \delta = 0$ does not reduce the support function, and Cauchy–Schwarz yields

$$\max_{\mathbf{1}^\top \delta = 0,\ \|\delta\|_2 \le \rho} (-\delta^\top a_t) = \rho \left\| a_t \right\|_2 , \tag{55}$$

attained at $\delta^\star = -\rho a_t / \left\| a_t \right\|_2$ when $a_t \ne 0$. This proves

$$S_t^{\mathrm{fo}}(\eta, \rho) = L_{V,t}(\theta_t) - \eta \langle g_{V,t}, P_t \bar{g}_t \rangle + \eta \rho \left\| a_t \right\|_2 . \tag{56}$$

For the covariance form,

$$\left\| a_t \right\|_2^2 = \sum_{i=1}^{m} \langle g_{V,t}, P_t (g_{i,t} - \bar{g}_t) \rangle^2 \tag{57}$$

$$= g_{V,t}^\top P_t \left(\sum_{i=1}^{m} (g_{i,t} - \bar{g}_t)(g_{i,t} - \bar{g}_t)^\top \right) P_t g_{V,t} \tag{58}$$

$$= m\, g_{V,t}^\top P_t C_t P_t g_{V,t}. \tag{59}$$

Substituting into the first expression proves the equivalent covariance form. □

A.3 Proof of Theorem 2

Proof. Fix $\alpha \in \mathcal{A}_\rho$ and let $\Delta(\alpha) = -\eta P_t g_{U,t}(\alpha)$. By Assumption 1,

$$L_{V,t}(\theta_t + \Delta(\alpha)) \leq L_{V,t}(\theta_t) + \langle g_{V,t}, \Delta(\alpha)\rangle + \frac{\beta}{2}\|\Delta(\alpha)\|_2^2 . \tag{60}$$

Since $\Delta(\alpha) = -\eta P_t g_{U,t}(\alpha)$,

$$L_{V,t}(\theta_t(\alpha)) \leq L_{V,t}(\theta_t) - \eta \langle g_{V,t}, P_t g_{U,t}(\alpha)\rangle + \frac{\beta\eta^2}{2}\|P_t g_{U,t}(\alpha)\|_2^2 . \tag{61}$$

Taking the maximum over $\alpha \in \mathcal{A}_\rho$ gives

$$S_t^{\mathrm{ex}}(\eta,\rho) \leq S_t^{\mathrm{fo}}(\eta,\rho) + \frac{\beta\eta^2}{2}\max_{\alpha\in\mathcal{A}_\rho}\|P_t g_{U,t}(\alpha)\|_2^2 . \tag{62}$$

Write $\alpha = u_m + \delta$. Then

$$g_{U,t}(\alpha) = \bar{g}_t + G_t\delta, \tag{63}$$

where $G_t \in \mathbb{R}^{p\times m}$ has columns $g_{i,t} - \bar{g}_t$. Hence

$$\begin{aligned}
\|P_t g_{U,t}(\alpha)\|_2 &\leq \|P_t\bar{g}_t\|_2 + \|P_t G_t\delta\|_2 && (64)\\
&\leq \|P_t\bar{g}_t\|_2 + \|P_t G_t\|_{\mathrm{op}}\|\delta\|_2 && (65)\\
&\leq \|P_t\bar{g}_t\|_2 + \rho\|P_t G_t\|_{\mathrm{op}} . && (66)
\end{aligned}$$

It remains to identify $\|P_t G_t\|_{\mathrm{op}}$. Because P_t is symmetric,

$$\|P_t G_t\|_{\mathrm{op}}^2 = \lambda_{\max}(G_t^\top P_t^2 G_t). \tag{67}$$

The nonzero eigenvalues of $G_t^\top P_t^2 G_t$ equal those of $P_t G_t G_t^\top P_t$. Since $G_t G_t^\top = mC_t$, we obtain

$$\|P_t G_t\|_{\mathrm{op}}^2 = m\,\lambda_{\max}(P_t C_t P_t) = \kappa_t(P_t)^2. \tag{68}$$

Therefore

$$\max_{\alpha\in\mathcal{A}_\rho}\|P_t g_{U,t}(\alpha)\|_2 \leq \|P_t\bar{g}_t\|_2 + \rho\kappa_t(P_t), \tag{69}$$

and substituting this into the previous inequality proves Eq. (23). □

A.4 Proof of Corollary 1

Proof. By Theorem 2,

$$\begin{aligned}
S_t^{\mathrm{ex}}(\eta,\rho) \leq\ & L_{V,t}(\theta_t) - \eta\langle g_{V,t}, P_t\bar{g}_t\rangle + \eta\rho\sqrt{m\, g_{V,t}^\top P_t C_t P_t g_{V,t}}\\
&+ \frac{\beta\eta^2}{2}\left(\|P_t\bar{g}_t\|_2 + \rho\kappa_t(P_t)\right)^2 . && (70)
\end{aligned}$$

The stated inequality makes the right-hand side strictly less than $L_{V,t}(\theta_t)$, which proves the claim. □

A.5 Proof of Proposition 2

Proof. By the strong law of large numbers,

$$\begin{aligned}
\hat{L}_{V,n}(\theta) &\to L_V(\theta) \quad \text{a.s.}, && (71)\\
\hat{\mu}_m(\theta) &\to \mu(\theta) \quad \text{a.s.}, && (72)\\
\hat{\nu}_n(\theta) &\to \nu(\theta) \quad \text{a.s.} && (73)
\end{aligned}$$

as $m, n \to \infty$. Likewise, each entry of the empirical second-moment matrix

$$\frac{1}{m}\sum_{i=1}^{m}\nabla\ell(z_i;\theta)\nabla\ell(z_i;\theta)^\top \to \mathbb{E}\left[\nabla\ell(z;\theta)\nabla\ell(z;\theta)^\top\right] \quad \text{a.s.} \tag{74}$$

Since

$$\hat{\Sigma}_m(\theta) = \frac{1}{m}\sum_{i=1}^{m} \nabla\ell(z_i;\theta)\nabla\ell(z_i;\theta)^\top - \hat{\mu}_m(\theta)\hat{\mu}_m(\theta)^\top, \tag{75}$$

we obtain

$$\hat{\Sigma}_m(\theta) \to \Sigma(\theta) \quad \text{a.s.} \tag{76}$$

Matrix multiplication and the square-root map are continuous on the cone of positive semidefinite matrices, so

$$\hat{\nu}_n(\theta)^\top P\hat{\Sigma}_m(\theta)P\hat{\nu}_n(\theta) \to \nu(\theta)^\top P\Sigma(\theta)P\nu(\theta) \quad \text{a.s.} \tag{77}$$

Finally, the map

$$(x, y, z, w) \mapsto x - \eta y^\top Pz + \eta\sqrt{\varepsilon\, y^\top PwPy} \tag{78}$$

is continuous on the relevant domain, so the continuous mapping theorem yields

$$\hat{\mathcal{S}}^{\text{fo}}_{m,n}(\theta) \to \mathcal{S}^{\text{pop}}_{\varepsilon}(\theta) \quad \text{a.s.} \tag{79}$$

□

A.6 Proof of Theorem 3

Proof. We prove the three parts in order.

Part (i). For family A,

$$\frac{1}{2}\left(\ell_1^A + \ell_2^A\right)(\theta + \Delta) = c + r^\top\Delta + \frac{\beta}{2}\|\Delta\|_2^2. \tag{80}$$

For family B,

$$\frac{1}{2}\left(\ell_1^B + \ell_2^B\right)(\theta + \Delta) = \frac{1}{2}\left[c + (1+\xi)r^\top\Delta + \frac{\beta}{2}\|\Delta\|_2^2 + c + (1-\xi)r^\top\Delta + \frac{\beta}{2}\|\Delta\|_2^2\right] \tag{81}$$

$$= c + r^\top\Delta + \frac{\beta}{2}\|\Delta\|_2^2. \tag{82}$$

Thus the pooled support loss, gradient, and Hessian are identical.

Part (ii). At $\Delta = 0$, family A has support gradients

$$g_1^A = g_2^A = r, \qquad \bar{g}^A = r. \tag{83}$$

Therefore $a^A = 0$ and

$$S_A^{\text{fo}}(\eta, \rho) = \ell_V - \eta\|r\|_2^2. \tag{84}$$

Family B has

$$g_1^B = (1+\xi)r, \qquad g_2^B = (1-\xi)r, \qquad \bar{g}^B = r. \tag{85}$$

Hence

$$a_1^B = \langle r, (g_1^B - \bar{g}^B)\rangle = \xi\|r\|_2^2, \qquad a_2^B = -\xi\|r\|_2^2. \tag{86}$$

So

$$\left\|a^B\right\|_2 = \sqrt{2}\,\xi\|r\|_2^2. \tag{87}$$

Applying Theorem 1,

$$S_B^{\text{fo}}(\eta, \rho) = \ell_V - \eta\|r\|_2^2 + \eta\rho\sqrt{2}\,\xi\|r\|_2^2, \tag{88}$$

which proves Eq. (41).

Part (iii). Any $\alpha \in \mathcal{A}_\rho$ can be written as

$$\alpha = \left(\frac{1}{2} + s,\ \frac{1}{2} - s\right), \qquad |s| \le \frac{\rho}{\sqrt{2}}. \tag{89}$$

For family A, the weighted support gradient is always r, so

$$S_A^{\text{ex}}(\eta, \rho) = L_V(\theta - \eta r) = \ell_V - \eta \|r\|_2^2 + \frac{\gamma\eta^2}{2} \|r\|_2^2 . \tag{90}$$

For family B, the weighted support gradient becomes

$$g_U^B(\alpha) = \left(\frac{1}{2} + s\right)(1+\xi)r + \left(\frac{1}{2} - s\right)(1-\xi)r \tag{91}$$

$$= (1 + 2\xi s)r. \tag{92}$$

Hence

$$L_V(\theta - \eta g_U^B(\alpha)) = \ell_V - \eta(1 + 2\xi s) \|r\|_2^2 + \frac{\gamma\eta^2}{2}(1 + 2\xi s)^2 \|r\|_2^2 . \tag{93}$$

Differentiate with respect to s:

$$\frac{d}{ds} L_V(\theta - \eta g_U^B(\alpha)) = 2\xi\eta \|r\|_2^2 \left(-1 + \gamma\eta(1 + 2\xi s)\right). \tag{94}$$

If Eq. (42) holds, then for all feasible s,

$$\gamma\eta(1 + 2\xi s) \le \gamma\eta(1 + \sqrt{2}\xi\rho) < 1, \tag{95}$$

so the derivative is strictly negative. Therefore the maximum occurs at the smallest feasible s, namely $s^\star = -\rho/\sqrt{2}$. Substituting this value gives

$$S_B^{\text{ex}}(\eta, \rho) = \ell_V - \eta(1 - \sqrt{2}\xi\rho) \|r\|_2^2 + \frac{\gamma\eta^2}{2}(1 - \sqrt{2}\xi\rho)^2 \|r\|_2^2 . \tag{96}$$

Subtracting $S_A^{\text{ex}}(\eta, \rho)$ yields

$$S_B^{\text{ex}}(\eta, \rho) - S_A^{\text{ex}}(\eta, \rho) = \sqrt{2}\xi\rho\eta \|r\|_2^2 + \frac{\gamma\eta^2}{2}\left[(1 - \sqrt{2}\xi\rho)^2 - 1\right] \|r\|_2^2 \tag{97}$$

$$= \sqrt{2}\xi\rho\eta \|r\|_2^2 (1 - \gamma\eta) + \gamma\xi^2\rho^2\eta^2 \|r\|_2^2 . \tag{98}$$

The right-hand side is strictly positive because Eq. (42) implies $\eta < 1/\gamma$. □

A.7 Proof of Proposition 3

Proof. Let $t^\star \in \arg\min_{t\in[0,T]} s(t)$. Choose a checkpoint t_k such that $|t_k - t^\star| \le \Delta/2$, which exists by construction of the grid. By Lipschitz continuity,

$$s(t_k) \le s(t^\star) + L_s|t_k - t^\star| \le s(t^\star) + \frac{L_s\Delta}{2}. \tag{99}$$

Taking the minimum over k proves the claim. □

A.8 Proof of Proposition 4

Proof. Since $Q \succeq 0$, the function

$$U(\theta) = c + b^\top\theta + \frac{1}{2}\theta^\top Q\theta \tag{100}$$

is convex. Jensen's inequality therefore gives

$$U\left(\sum_{t\in W} w_t\theta_t\right) \le \sum_{t\in W} w_t U(\theta_t) \tag{101}$$

for any nonnegative weights summing to one. □

B Additional Notes for Implementation

Exact inner maximization. The exact score in Eq. (8) is a low-dimensional smooth maximization problem over the simplex neighborhood $\mathcal{A}_\rho$. In practice, projected gradient ascent over the weight vector α is sufficient because the inner problem dimension is the size of the support subset, not the parameter dimension.

Diagonal preconditioning. The default theory uses a symmetric positive semidefinite P_t. A practical diagonal choice is

$$P_t = \mathrm{diag}(\hat{v}_t + \tau)^{-1/2}, \tag{102}$$

where $\hat{v}_t$ is the optimizer second-moment state or a diagonal empirical Fisher and $\tau > 0$ is a damping term.

Why the method is label-free. Neither the score nor the theory uses source-domain identifiers. The ambiguity set is defined over support-example weights, so the method can be applied wherever one has differentiable losses and held-out support/validation splits.

What must be implemented carefully. The main numerical sensitivity is the projected covariance term

$$g_{V,t}^\top P_t C_t P_t g_{V,t}. \tag{103}$$

If per-example gradients are very noisy, support subsampling or averaged micro-batches should be used to stabilize its estimation.